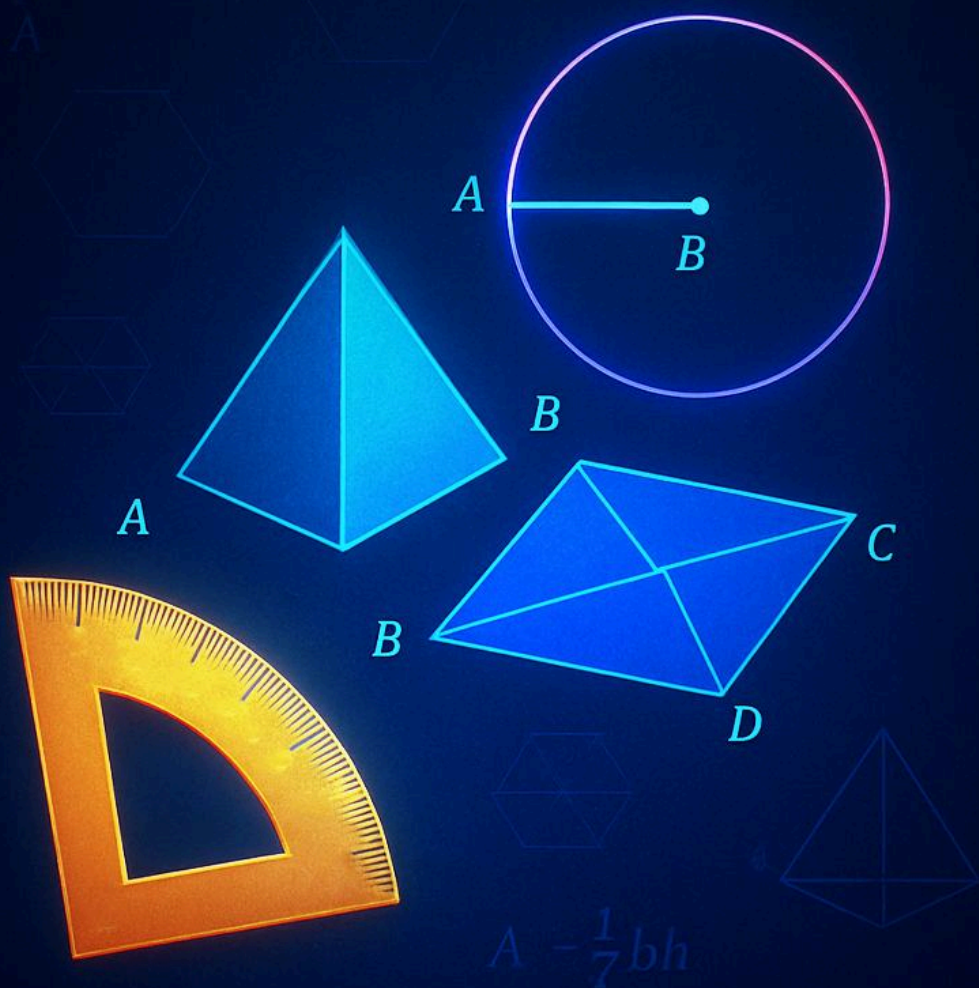


USEFUL FORMULAE & CONCEPT

GEOMETRY

CAT QUANTITATIVE APTITUDE



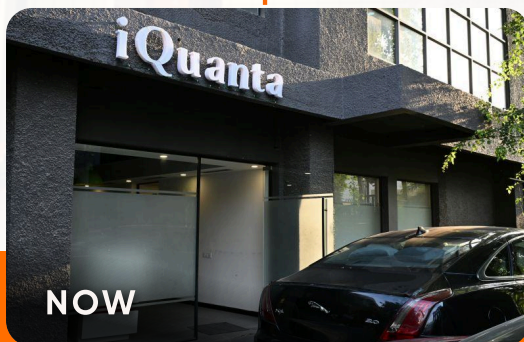
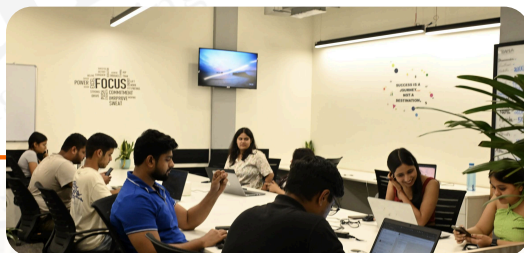
About iQuanta?

Founded in 2015 by Indrajeet Singh, iQuanta began as a Facebook doubt-solving group from a small apartment. With strong expertise in Quants, he earned the title 'Wizard of Quant' by solving thousands of students' queries with clarity and speed. Noticing the lack of quality guidance for CAT aspirants, he officially launched iQuanta to bridge this gap.

Over the years, iQuanta has continued to support over 4 lakh students through its active online communities, offering resources, and peer-driven learning. In the last 5 years, more than 8,000 students have converted IIMs. In 2024, iQuanta produced 500+ 99 percentilers, many of whom made it to top IIMs. iQuanta continues to guide thousands of aspirants on their journey to top B-schools.



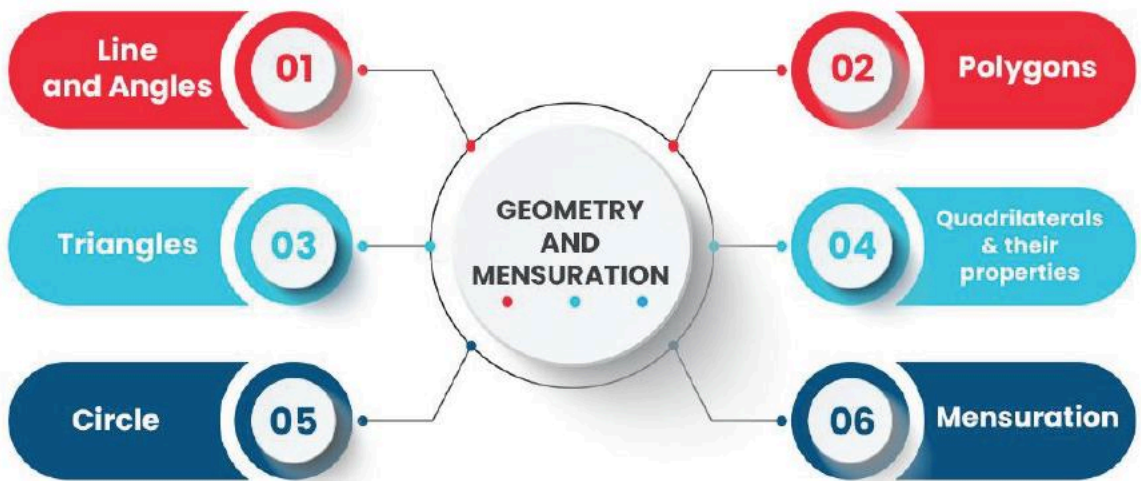
5 YEARS AGO



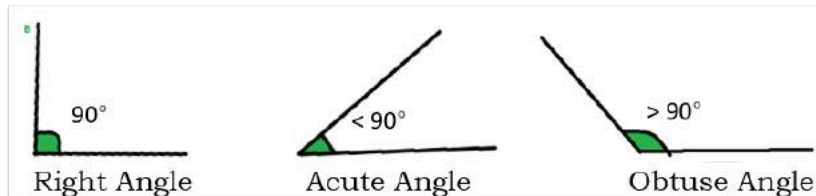
NOW



Indrajeet Singh
Founder & CEO, iQuanta
Forbes 30 Under 30



GEOMETRY

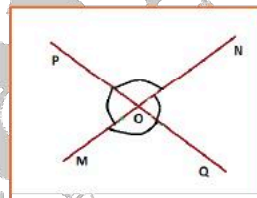


Complementary & Supplementary Angles:

Two angles which add up to 90 degrees are called complementary angles and two angles which add up to 180 degrees are called supplementary angles.

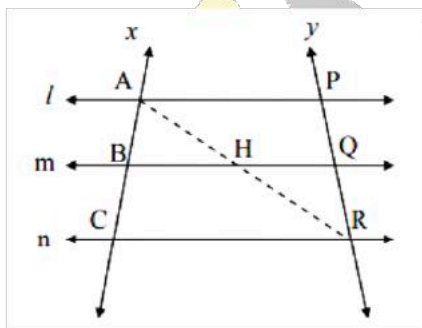
Vertically Opposite Angles:

$$\angle POM = \angle NOQ \text{ \& } \angle PON = \angle MOQ$$



Point to be remember:

The ratio of intercepts formed by a transversal intersecting three parallel lines is the same as the ratio of intercepts formed by any other transversal.



$$i.e., \quad \frac{AB}{BC} = \frac{PQ}{QR}$$

Polygons:

Straight sided, 2-D shapes that close in a space are known as polygons.

- Number of diagonals in an n-sided polygon = $\frac{n(n-3)}{2}$
- Sum of all the exterior angles of any polygon = 360°
- Measure of each exterior angle of a regular polygon = $\frac{360^\circ}{n}$
- Sum of all interior angles of any polygon = $(n-2) \times 180^\circ$
- Measure of each interior angle in a regular polygon = $\frac{(n-2)180^\circ}{n}$



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Triangle:

Type of triangles	Characteristics	Figure
Scalene Triangle	A triangle with sides of differing lengths.	
Isosceles Triangle	Is one with two sides that have the same length.	
Equilateral Triangle	A triangle with all sides of equal length	
Right-angled Triangle	One angle of 90°	
Obtuse-angled Triangle	One of the angles of the triangle is more than 90°	
Acute-angled Triangle	All the angles of the triangle are less than 90°	
Isosceles Right-angled Triangle	A right-angled triangle, whose two sides containing the right angle are equal in length.	

Pythagoras Theorem:

Pythagoras theorem is applicable in the case of a right-angled triangle. It says that the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$\text{Hypotenuse}^2 = \text{Base}^2 + \text{Perpendicular}^2$$

700+ 99%ilers & 1000s of CAT Toppers



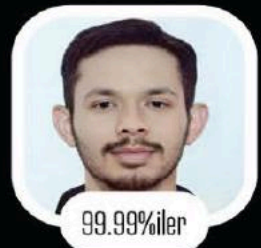
Rishi Mittal



Sabyasachi



Mridul Agarwal



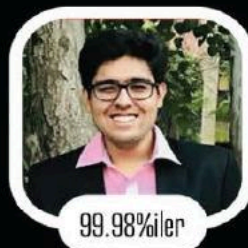
Parikshit Tomar



Guru Charan



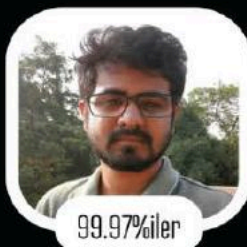
Manav Jain



Shikhar Sachdeva



Mihir Kapse



Satyajit Das



Swadesh Rath



Kislay Jha



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Topper's Interview

iQuanta = Relevant Content!



Another CAT 22 Slot 1 same & simple question done in iQuanta Course ...

Q. 20 identical chocolate to be distributed among 4 children such that each get some chocolate and they don't get odd number of chocolate

Soln : Not Odd means even $\Rightarrow 2a + 2b + 2c + 2d = 20$

$$a + b + c + d = 10$$

Now each gets some so natural soln $\Rightarrow (10-1)C(4-1) = 9C3$



QA: 49c3

Soln: As all even numbers are multiples of 2, so we can express all as even numbers that is

Let's take $x = 2a, y = 2b, z = 2c, w = 2d$

$$\text{so } 2a + 2b + 2c + 2d = 100,$$

$$a + b + c + d = 50,$$

so its natural solutions will be

$$(50-1)C(4-1) = 49C3$$

Clear ?

Amrat Sharma and 217 others

27 comments



Q4. Number of even natural solutions of $x+y+z+w = 100$ is ?

Amrat Sharma and 152 others

64 comments



Inika Singh

I was so worried about quants but almost every single question was something I had already practised several times in class, so relieved 😊

7m Love Reply



Mohammad Omar Farooq

Ek rc karva rakha tha practice mai Moderate tha 3-4 question gk ke iift ke gk session mai the

2 h Love Reply



Manish Khandelwal

Parajumble Competitive advantage wala which I have seen before in course

Just now Like Reply



Manish Khandelwal

Stoicism RC yeh v pehle dekha hai

Just now Like Reply



CAT Preparation - iQuanta

Jeet Singh · 22 h · 📷

#iQuanta250_QA Series by INDRA

Q132. PNC !

CAT Preparation - iQuanta

iQuanta 250 Series

In how many ways can 7 chocolates of type A, 6 chocolates of type B and 5 chocolates of type C be distributed among 4 children such that each child receives at least one chocolate of each type?



Yeh waala Aaya tha slot 2 me



Shreya Rathi

Kafi quant ka questions match kia and Irdi ka ek set to pura same class me krwaya tha...Lrdi k kafi similar questions... Thankyou iQuanta team..

Jeet Singh Jainam Shah Yogesh Joshi Hitesh Hariramani ❤️❤️❤️

43 m Love Reply



Properties of a triangle:

- The sum of all the angles of a triangle = 180° ($\angle a + \angle b + \angle c = 180$)
- The sum of lengths of any two sides > length of the third side
- The difference of any two sides of any triangle < length of the third side

The area of any triangle formula:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = r \times s = (a \times b \times c) / 4R = \frac{1}{2} \times a \times b \times \sin C$$

where a, b and c are the sides of the triangle, r is the inradius, R is circumradius, and s is the semi perimeter.

By Heron's formula, the area of the triangle is given by:

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

where 's' is the semi perimeter

$$s = \frac{a + b + c}{2}$$

Some important Definitions:

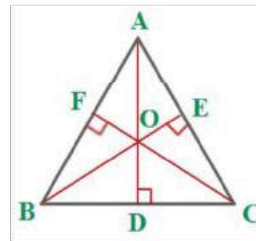
Altitude of Triangle: Perpendicular drawn from a vertex of a triangle to the side opposite side.

Median of triangle: The straight lines drawn from the vertex of triangle that bisect the opposite sides of the triangles is called median of a triangle.



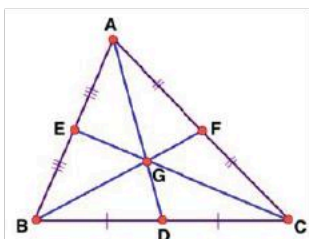
Orthocentre: The point inside a triangle where the altitudes meet is called orthocentre of the triangle.

AD, BE and CF are altitudes of the triangle and O is the orthocentre of the triangle.



Circumcentre: The point inside a triangle where the perpendicular bisectors of each side meet.

Centroid: The place/point where all the medians meet is called centroid of the triangle. The Centroid divides each median in the ratio 2: 1.



AD, BE, CF – medians of $\triangle ABC$.

G is the centroid of the $\triangle$.

AG: GD = 2: 1

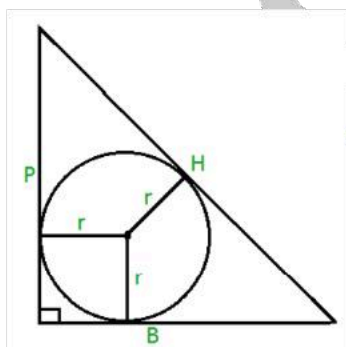
CG: GE = 2: 1

BG: GF = 2: 1

Centroid divides the line joining the circumcentre & orthocentre in ratio 2:1.

Incentre: The point inside a triangle where the angle bisectors meet.

Note: All the above points are same in case of Equilateral triangle.



Right angle triangle:

$$\text{Circumradius} = \frac{H}{2}$$

$$\text{Inradius (r)} = \frac{P+B-H}{2}$$

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99.98%iler



Swaraj Pal
99.95%iler



Raunak Das
99.55%iler



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Mridul Tiwari
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Aman Ehtesham
99.94%iler

IIM Calcutta Converts



Mridul Tiwari
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Sanvie Singhal
99.92%iler



Chitturi Sakshvith
Sanjeev
99.86%iler



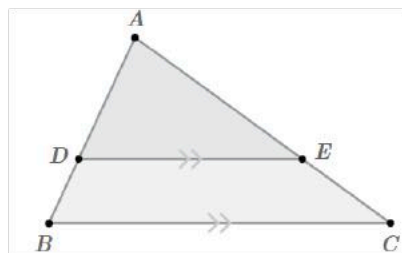
Madhumitha
Sriharsha
99.6%iler

And many more

Important Theorems for Triangles

Basic Proportionality Theorem (BPT):

Any line parallel to one side of a triangle divides the other two sides proportionally. So, if DE is drawn parallel to BC, then it would divide sides AB and AC proportionally.

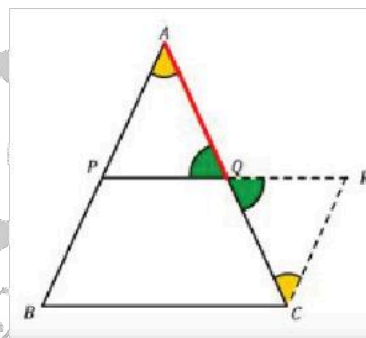


i.e., $AD/DB = AE/EC$

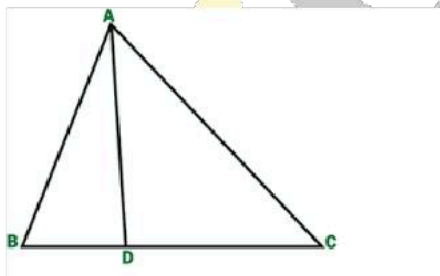
Mid-point theorem: The line segment joining mid-points of two sides of a triangle is parallel to the third side of the triangle and is half of it.

P and Q are the mid points of AB and AC respectively.

$PQ \parallel BC$ & $PQ = BC/2$



Apollonius' theorem: In a triangle, the sum of the squares of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side.



$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

(AD is median)

Journey from 51% in Grads to converting IIM Ranchi, Trichy & 8 other IIMs.



Prateek Gupta

1 hr · 🌐

"Have the courage to follow your heart and intuition. They somehow know what you truly want to become" - Steve Jobs

Back in 2018, when I graduated from the University of Delhi with a mere 51% in English Literature, I felt like my life was over. I somehow managed to get a basic job at a big company and worked 13 hours a day sometimes. But I realized that an arts degree limits you to grow in a face-paced organization. That was when I decided to go for CAT in 2019. I started self-study and somehow managed to diligently put 2 hours daily. But my strategy was so random and without any guidance that ultimately resulted in a 76 percentile.

And that's when I decided to switch to a more lenient job, a decision which everyone around me graciously rejected. But the strategy was still random and chaotic. When there were only 4 months left to the D-day, I decided to join iQuanta. It provided me everything I needed: Regular classes, 24/7 doubt solving, updated portal (new pattern mocks), and the most important thing - very helpful and kind Indrajeet sir. Moreover, I bought a white-board and went berserk over every doubt posted on the group to prepare for my final attempt.

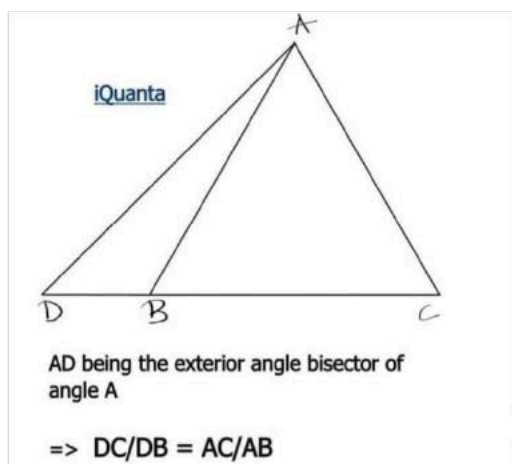
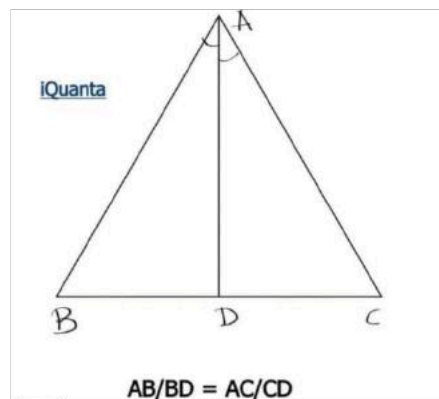
But when life gives you a lemon, you find the best possible to deal with it. I was having a 100-degree fever on D-Day and somehow managed the paper sitting in an isolated portion of the examination hall. Honestly, I lost all hopes after that. I started focusing on my job and was completely astonished on 2nd Jan. I scored a 95.45 percentile and got calls from IIM Rohtak, CAP (IIM Trichy, Ranchi, etc), Vizag, IIM ND, and Nagpur.

After the interviews, the same astonishment struck me yesterday when I cleared all the IIM CAP colleges in the first list. And while writing this post, I converted IIM Rohtak as well. I am at a loss of words for the seamless support and guidance that iQuanta offered. The best thing you can take from iQuanta is the positivity that is departed from the people teaching you. I know it's hard for each and every aspirant, but with proper guidance and hard work you can push your past mistakes. I wish that everyone who is reading this converts their dream b-school. Also, I will be glad to help in any way possible. All the best and remember that you can't work on the past but only on a better future.



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Interior angle Bisector theorem: In a triangle, the angle bisector of an angle divides the opposite side to the angle in the ratio of the remaining two sides.



Exterior angle Bisector theorem: In a triangle, the angle bisector of any exterior angle of a triangle divides the side opposite to the external angle in the ratio of the remaining two sides

Similarity of Triangles:

- AA similarity (angle – angle)
- SSS similarity (side – side – side)
- SAS similarity (side – angle – side)

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Types of Quadrilaterals

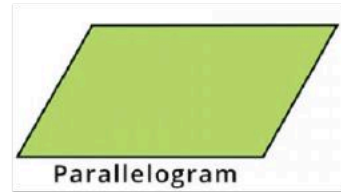
Quad = four. So, quadrilaterals are polygons with 4 sides. Square & Rectangle are the most basic quadrilaterals.

Parallelogram: A parallelogram is a quadrilateral when its opposite sides are equal and parallel. The diagonals of a parallelogram bisect each other.

Perimeter = $2 \times \text{sum of adjacent sides}$

Area = $b \times h$

(b – width, h – length of perpendicular)

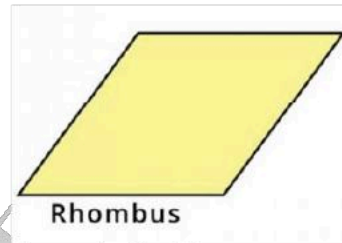


Rhombus: A rhombus is a quadrilateral when all sides are equal. The diagonals of a rhombus bisect each other at right angles (90°)

Perimeter = $4 \times \text{Side}$

Area = $\frac{1}{2} \times d_1 \times d_2$

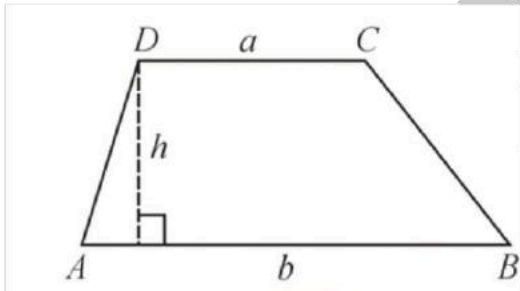
(d_1 and d_2 are diagonals)



Trapezium: A trapezium is a quadrilateral in which only one pair of the opposite sides is parallel.

Area = $\frac{1}{2} \times (a + b) \times h$

Perimeter = Sum of all four sides



CIRCLE:

Area = Πr^2

Perimeter = $2\Pi r = \Pi d$ (as we know $d = 2r$)

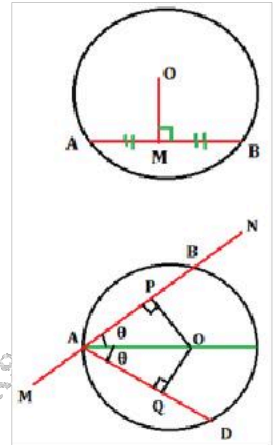
Properties of circles:

1. $\perp$ bisector of any chord always passes through centre & vice – versa is also true.

2. If the $\perp$ distances from the centre are equal then chords are equal

If $OP = OQ$, then $AB = AD$

(MN = secant, AB & AD are chords. Arc AB = minor arc,
Arc ADB = major arc.)



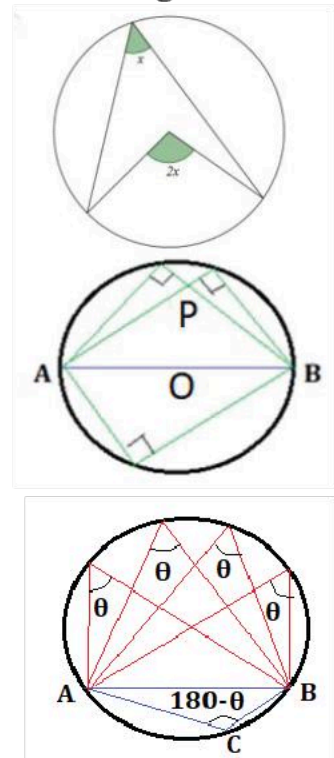
3. The angle bisector of two equal chords always passes through the centre.

4. The angle at the center of a circle is twice the angle at the circumference.

a) So, angle made by the diameter on the circumference = $180^\circ/2 = 90^\circ$

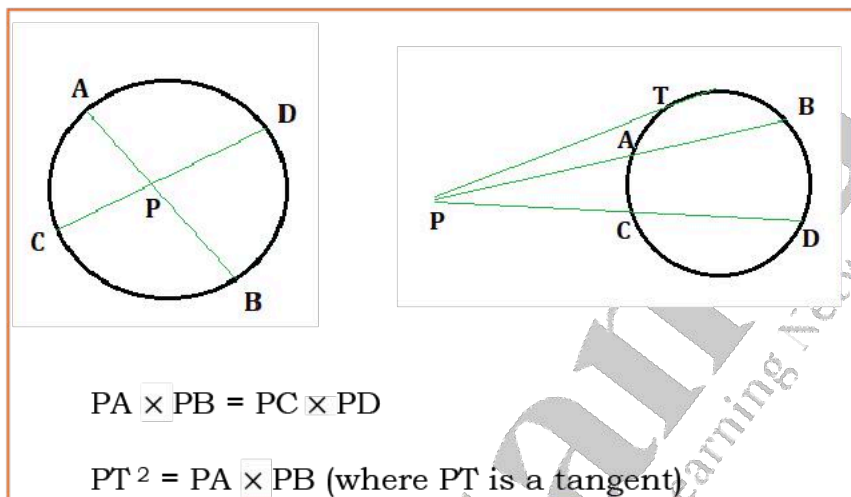
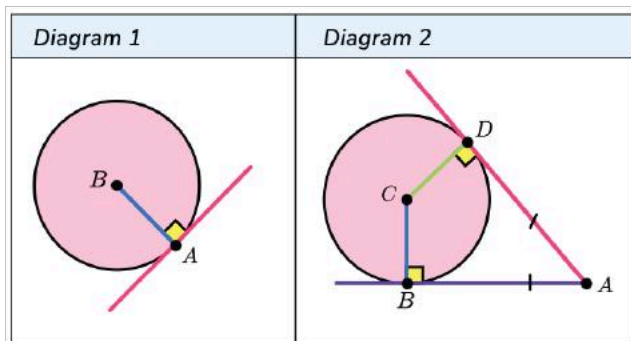
b) Angle made by a chord in the major arc is & in minor arc is obtuse & these angles are supplementary.

c) Angles made by a chord or chords of equal lengths on the circumference are equal.



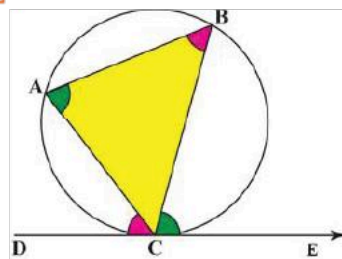
Tangent of circle: A straight line that touches the circumference of the circle at only one point.

- The angle between a tangent and radius is 90 degrees.
- Tangents drawn by a point on circle are always equal. $AB = AD$



Alternate segment theorem:

For any circle, the angle formed between the tangent and the chord through the point of contact of the tangent is equal to the angle formed by the chord in the alternate segment



Mensuration:

Cube:

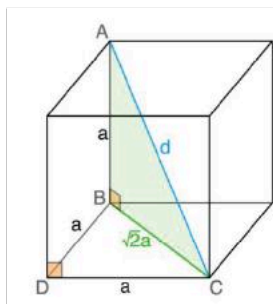
$$V = a^3$$

$$L.S.A = 4a^2$$

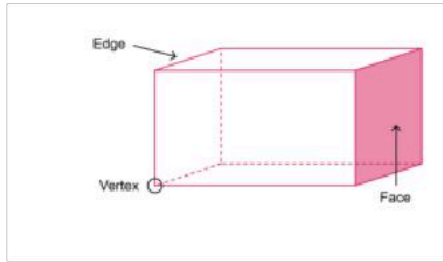
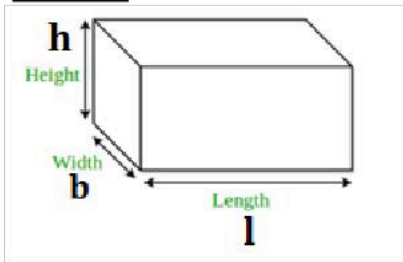
$$T.S.A = 6a^2$$

$$\text{Length of each face diagonal} = a\sqrt{2}$$

$$\text{Length of body diagonal} = a\sqrt{3}$$



Cuboid:



$$V = l.b.h$$

$$LSA = 2 (lh + bh) = 2h (l + b)$$

$$TSA = 2 (lb + bh + hl)$$

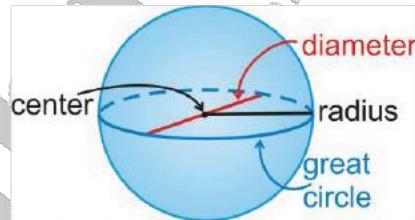
Lengths of face diagonals are $\sqrt{l^2 + b^2}$, $\sqrt{l^2 + h^2}$, $\sqrt{b^2 + h^2}$

Length of body diagonal = $\sqrt{l^2 + b^2 + h^2}$

Sphere:

$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$\text{Total surface area} = 4 \pi r^2$$



Hemisphere:

$$\text{Volume} = \frac{2}{3} \pi r^3$$

$$\text{C.S.A} = 2 \pi r^2$$

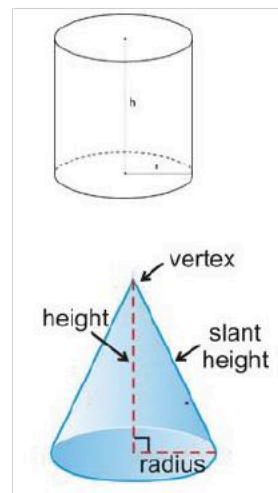
$$\text{T.S.A} = 3 \pi r^2$$

Cylinder:

$$V = \pi r^2 h$$

$$LSA = 2 \pi r h$$

$$TSA = 2 \pi r^2 + 2 \pi r h = 2 \pi r (r + h)$$



Cone:

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$LSA = \pi r l$$

$$TSA = \pi r^2 + \pi r (r + l)$$

$$\text{Slant height} = \sqrt{r^2 + h^2}$$

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CAT Course

COURSE FEATURES	CAT FULL COURSE	CAT FULL COURSE PRO	OTHER COACHINGS
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Assignments	✓	✓	✓
24*7 Doubt Solving	✓	✓	✗
Full Mocks with Video Solution	✓	✓	✓
Sectionals with Video Solution *	✓	✓	✗
Non-Engineer's Quant *	✓	✓	✗
Engineer's VARC *	✓	✓	✗
LRDI Inception *	✓	✓	✗
Ai Mock Analysis	✓	✓	✗
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12 Sets: CAT Level Books *	✗	✓	✗

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9 Times CAT,
VARC 99%iler

Shabana Ma'am

CAT VARC
99.5+%iler

